EXPERIMENT 01

GPT Language Model Training Baseline Report

WikiText-103 (5,000 rows) | ~300K Tokens | GPT-2 BPE Tokenizer

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Dataset	Hardware	Date	Epochs
WikiText-103 (~300K tokens)	Google Colab T4	May 2026	5

1. Objective

Experiment 01 established the initial GPT decoder-only transformer baseline using a compact WikiText-103 subset. The experiment focused on validating the training pipeline, tokenizer integration, checkpoint saving, and language model training behaviour on a small-scale corpus.

2. Model Configuration

The model architecture follows a standard decoder-only transformer design with tied embedding weights. This compact configuration produced approximately 16 million parameters and served as the initial baseline before scaling experiments.

Component	Configuration	Parameters (Approx.)
Token Embedding	50,257 × 256	12.9 M
Position Embedding	128 × 256	0.03 M
Transformer Layers	4 Layers | d_model=256 | heads=4	~3.1 M
LM Head	Tied Weights	—
Total	Decoder-only Transformer	~16 M

3. Hyperparameter Configuration

Parameter	Value
Vocabulary Size	50,257
Context Length	128
d_model	256
Attention Heads	4
Transformer Layers	4
Dropout	0.1
Learning Rate	3e-4
Minimum Learning Rate	1e-5
Gradient Accumulation	4 Steps
Mixed Precision (AMP)	Enabled

4. Training Setup

Setting	Value
Optimizer	AdamW
Weight Decay	0.1
Gradient Clipping	1.0
Learning Rate Schedule	Cosine Decay
Sliding Window Strategy	Highly Overlapped
Checkpoint Strategy	latest.pt only
Flash Attention	Not Used

5. Training Results

Epoch	Train Loss	Validation Loss	Perplexity
1	3.4662	8.6967	5,983
2	0.9319	10.7294	45,681
3	0.4473	11.5839	107,358
4	0.3232	11.8878	145,483
5	0.2834	11.9652	157,185

6. Failure Diagnosis

Training loss decreased consistently while validation loss increased sharply across epochs, indicating severe overfitting. The model memorized the training corpus instead of learning generalized language representations. Validation perplexity increased from 5,983 to 157,185, confirming poor generalization behaviour.

7. Root Cause Analysis

Factor	Severity	Observation
Dataset Size	High	300K tokens insufficient for ~16M parameters
Sliding Window Overlap	High	Reduced effective data diversity
Checkpoint Strategy	High	Best checkpoint overwritten
Regularization	Medium	Dropout insufficient for dataset scale
Flash Attention	Low	No impact on generalization

8. Improvements Planned for Experiment 02A

Experiment 02A addressed the major limitations identified in Experiment 01 by scaling the dataset to approximately 5 million medical-domain tokens, increasing model capacity to approximately 70 million parameters, reducing sliding-window overlap, and implementing proper checkpoint preservation using best.pt and epoch-wise checkpoints.

9. Conclusion

Experiment 01 successfully validated the GPT training pipeline while exposing major limitations in data scale, checkpoint management, and generalization behaviour. Although the model failed to generalize effectively, the experiment provided critical engineering insights that directly informed the successful redesign implemented in Experiment 02A.